Cross-Scene Hyperspectral Image Classification With Discriminative Cooperative Alignment

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***Abstract*— Cross-scene classification is one of the major challenges for hyperspectral image (HSI) classification, especially for target scenes without label samples. Most traditional domain adaptive methods learn a domain invariant subspace to reduce statistical shift while ignoring the fact that there may not exist a shared subspace when marginal distributions of source and target domains are very different. In addition, it is important for HSI classification to preserve discriminant information in the original space. To solve this issue, discriminative cooperative alignment (DCA) of subspace and distribution is proposed to cooperatively reduce the geometric and statistical shift. In the proposed framework, both geometrical and statistical alignments are considered to learn subspaces of the two domains with preserving discrimination information. Furthermore, a reconstruction constraint is imposed to enhance the robustness of subspace projection. Experimental results on three cross-scene HSI data sets demonstrate that the proposed DCA is significantly better than some state-of-the-art domain-adaptive approaches.**

***Index Terms*— Cross-scene, distribution alignment, domain adaption, hyperspectral image (HSI) classification, subspace alignment (SA).**

I. Introduction

IN RECENT years, sensors and image processing techniques have been developed rapidly, and acquisition of remote sensing (RS) data with the high spatial and spectral resolution has become easier. It brings us not only benefits but also many challenges. Since in traditional machine learning algorithms, such as support vector machine (SVM) [1], the sparse representation-based classifier (SRC) [2], and collaborative representation-based classifier (CRC) [3], labeling process usually takes a lot of time and human labor. Therefore, in a real hyperspectral image (HSI) classification application, training samples in the source scene are used to classify a newly acquired similar target scene with no labeled sample. This challenge is the so-called cross-scene task, which aims to transfer shared knowledge in the source scene in order to improve classification performance in the target scene. In other words, labeled classes with similar scenes can be utilized as training samples.

When directly using the source scene to classify the target scene, one of the problems on spectral shift is usually encountered. Due to different sensor nonlinearities, different substance compositions of the same land cover class in different sizes and times, and so on [4], the same ground objects from source domains (SD) and target domains (TD) differ in conditional distribution and spectral reflectance. For instance, Fig. 1 illustrates the conditional distribution and spectral reflectance of two hyperspectral data sets (e.g., HyRANK and Houston), where the HyRANK's Dioni (source scene) and the Loukia (target scene) data were obtained for the same sensor (i.e., Hyperion) for different scenes, and the Houston's 2013 (source scene) and 2018 (target scene) data were obtained for different sensors at different times. In Fig. 1, red and blue curves represent spectral signatures of the source data and the target data, respectively, and Fig. 1(a)–(c) are class 7 from the Houston data and others are class 10 from the HyRANK data. Therefore, domain adaptation (DA) is applied to cross-scene classification, whose setting is that the source task and the target task are the same while SD and TD are different but related [5]. DA is a case of transductive transfer learning, which can reduce the spectral shift in feature-level and learn domain-invariant models.

Traditional DA contains three types of methods: distribution alignment-based methods, feature selection-based methods, and subspace learning-based methods. The basic idea of distribution alignment-based methods is that since data probability distributions of SD and TD are different, a direct way is to shorten the distance of different data distributions through some transformations. According to the nature of the data distribution, the relevant methods can be divided into marginal distribution alignment, conditional distribution alignment, and joint distribution alignment [6]–[8]. The feature selection-based methods assume that both SD and TD contain part of common features, where the data distribution of SD

Manuscript received October 14, 2020; revised November 27, 2020; accepted December 15, 2020. Date of publication January 13, 2021; date of current version October 26, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61922013, Grant U1833203, and Grant 61772225; in part by the Beijing Natural Science Foundation under Grant JQ20021; in part by the National Natural Science Foundation of Guangdong under Grant 2018B030311046; and in part by the National Natural Science Foundation of China under Grant 61871177. *(Corresponding author: Wei Li.)*

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Digital Object Identifier 10.1109/TGRS.2020.3046756

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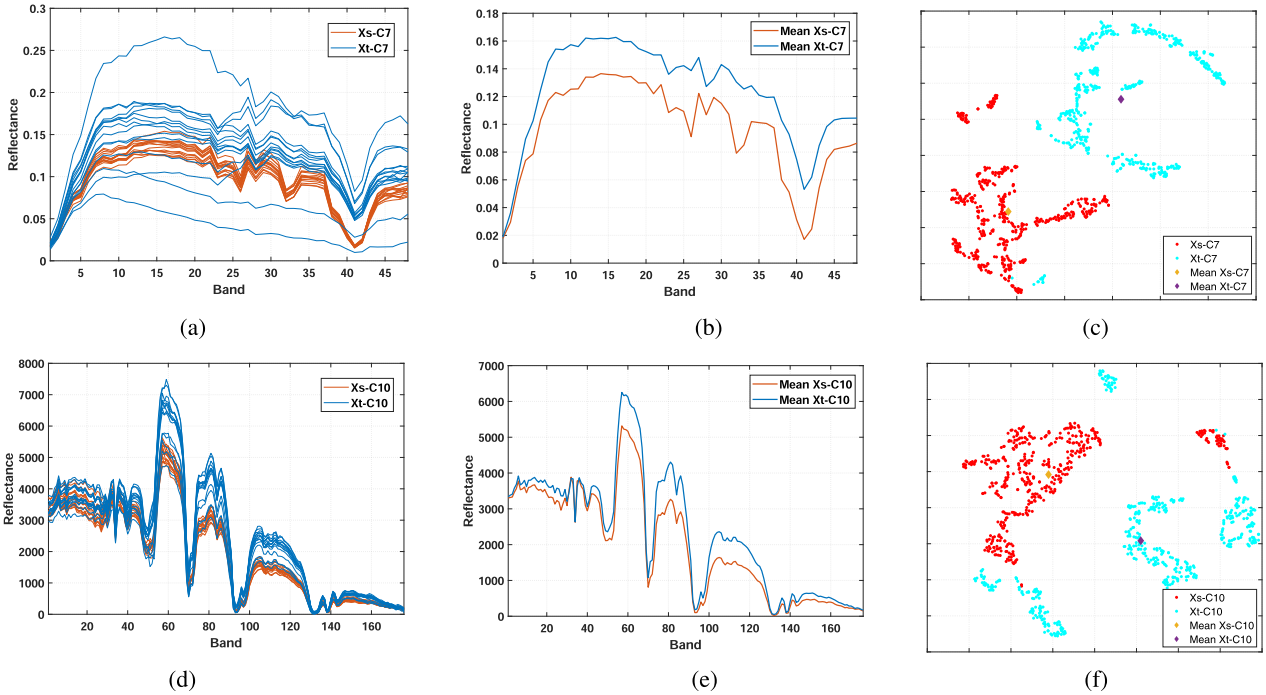


Fig. 1. Spectral reflectance and distribution visualization of two land-cover classes in the Houston and HyRANK data. (a)–(c) Houston data. (d)–(f) HyRANK data.

and TD are consistent. Therefore, the goal of such methods is to use machine learning methods to select shared features, and then build one model based on these features [9]–[11]. The subspace learning-based methods usually assume that SD and TD data have similar distributions in the transformed subspace. According to the form of feature transformation, they are divided into two categories, i.e., statistical feature alignment methods based on statistical feature transformation and manifold learning methods based on manifold transformation [12]–[14].

Recently, many researchers have examined the problem of DA for HSI classification, and DA techniques have been successfully applied to them. Sun *et al.* [15] introduced the domain transfer multiple-kernel learning to simultaneously minimize the maximum mean discrepancy (MMD) and the structural risk functional of SVM. Rotation-based ensemble and transfer component analysis (TCA) was proposed, where features in both source and target images were divided into several disjoint feature subsets [16]. In [17], semisupervised TCA was further developed, which was designed to statistically align the target data to the source data in reproducing kernel Hilbert space (RKHS). Both [18] and [19] iteratively labeled and added representative samples in TD to the training set in process of active learning. In [20], multiple linear transformations were employed to find new representations for different classes of samples from SD, and low-rank reconstruction was used to reduce the influence of outliers and noises. Subspace alignment (SA) was utilized to scene classification [21]. Luo *et al.* [22] applied the manifold regularization in conjunction with MMD to each class, minimizing the domain shift on a per-class basis and constrain data manifold of both source and target data to be preserved. A spatial-spectral multiple geodesic flow kernel learning (S2-MGFKL) [23] was presented to exploit spatial–spectral features and construct Gaussian GFKs to be fed into SVM.

Although many DA research studies have been carried out, and satisfied classification performance has been obtained, there are still three deficiencies that limit the application of DA methods in cross-scene HSI classification. First of all, almost all distribution alignment-based methods assume that there is only a common subspace that makes the distribution of SD and TD similar. However, when SD and TD are quite different, it is difficult to have a common subspace containing the shared features of the two domains. Therefore, the domain distribution is difficult to align in this case. Second, distribution adaptation-based methods only focus on aligning the distribution shift of projection samples of the two domains; that is, they can reduce the statistical spectral shift while ignoring the geometric spectral shift of the two domains, where the meaning of geometric shift is the shift of two domains on the bases of low-dimensional subspace after projection, that is, the spectral common basis of two domains has shift, and the meaning of statistical shift is the shift of the probability distribution (marginal distribution and conditional distribution) of the two domains in the spectral dimension. Third, many DA methods do not takeinto account that due to the high spectral dimension in HSI. When the hyperspectral data of SD and TD are projected to unrelated dimensions into a low-dimensional subspace, the discrimination information may be incomplete or even lost.

To solve the aforementioned three problems, both geometrical alignment and statistical alignment are considered to learn subspaces of SD and TD separately with preserving discrimination information ability. Discriminative cooperative

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alignment (DCA) of subspace and distribution as unsupervised DA is proposed in this article. The alignment of the two domains is performed from two perspectives. On the one hand, DCA applies a constraint to align the two subspaces, which can control the similarity of data geometric structure after projection, while maintaining the global geometric structure, so that the projection of SD and TD do not deviate too much. On the other hand, it calculates the interdomain MMD and intraclass MMD to align the marginal and conditional distribution of two projected domains. In addition, a data reconstruction constraint is imposed to ensure that the aligned data preserve most discriminant information.

The main contributions of this work are summarized as follows: 1) to deal with difficulties in cross-scene HSI classification, domain alignment is designed to reduce the statistical and geometric shift; 2) alignment of subspace and distribution is performed cooperatively to achieve better adaption performance; and 3) through imposing data reconstruction constraints, properties of source scene and target scene are retained to the greatest extent for learning a more robust feature subspace.

The rest of this article is organized as follows. Section II reviews the related methods of previous DA. In Section III, the theoretical framework and optimal solution of the proposed DCA are provided. The comparison of performance of the proposed method with several DA methods is present in Section IV. Finally, conclusions are drawn in Section V.

II. Related Work

A. Subspace Learning-Based Domain Adaptation

Subspace learning-based methods reduce the geometric domain shift by manipulating subspaces (such as building linear mapping or manifold learning) of SD and TD, so that subspaces of each domain contribute to the final mapping [12], [24], [25]. As one of the representative methods, SA [12] creates subspaces from the two domains, and learns a linear transformation $\mathbf{M}$ that aligns source subspace with target subspace, and realizes transformation and alignment of different source data. The optimization objective is described as

$$\min_{\mathbf{M}} \ \|\mathbf{W}_s\mathbf{M} - \mathbf{W}_t\|_F^2 \tag{1}$$

where $\mathbf{W}_s \in {}^{d\times k}$ and $\mathbf{W}_t \in {}^{d\times k}$ are bases of source and target subspaces, respectively, which are k eigenvectors corresponding to the k largest eigenvalues obtained via principal component analysis (PCA), and d is the dimension of source and target data. The closed-form solution of the above optimization problem can be directly obtained as

$$\mathbf{M}^* = \arg\min_{\mathbf{M}} \ \left\|\mathbf{M} - \mathbf{W}_s^T\mathbf{W}_t\right\|_F^2 = \mathbf{W}_s^T\mathbf{W}_t. \tag{2}$$

SA is simply implemented and its calculation is efficient, but only the first-order feature alignment of SD and TD is performed. Sun *et al.* [13] proposed the correlation alignment method (CORAL), which aligns the two domains with second-order features and learns a second-order feature transformation $\mathbf{A}$ to minimize the feature distance between the two domains

$$\min_{\mathbf{A}} \ \|\mathbf{A}^T\mathbf{C}_s\mathbf{A} - \mathbf{C}_t\|_F^2 \tag{3}$$

where $\mathbf{C}_s$ and $\mathbf{C}_t$ are covariance matrixes of the two domains, respectively. The solution to CORAL is also efficient, and it is widely employed as loss function of deep learning.

The DA based on manifold transformation is also one of the most popular methods, which employs manifold transformation to reduce the difference between separate domains. For example, in Grassmann manifolds, feature transformation and distribution adaptation usually have effective numerical forms, so they can be represented and solved very efficiently in DA problems [26]. Therefore, it is feasible to use the Grassmann manifold for DA. There are many existing DA methods to transform the original features into a manifold [14], [26]–[28].

B. Distribution Alignment-Based Domain Adaptation

Distribution alignment is one of the most commonly used DA methods. Assuming that $\mathbf{X}_s \in {}^{d\times n_s}$ and $\mathbf{X}_t \in {}^{d\times n_t}$ are data from SD and TD, respectively (here, d, n_s and n_t are dimension of data, number of source and target samples, respectively), $\mathbf{Y}_s$ and $\mathbf{Y}_t$ are the corresponding class labels, respectively (here, there is no $\mathbf{Y}_t$ in unsupervised case). The marginal distribution and conditional distribution are expressed as $P(\mathbf{X})$ and $P(\mathbf{Y}|\mathbf{X})$, respectively. TCA is a classical method of marginal distribution adaptation, which assumes that there is a feature map ϕ, so that distribution of the mapped data is approximate, i.e., $P(\phi(\mathbf{X}_s)) \approx P(\phi(\mathbf{X}_t))$. In addition, it is assumed that if the marginal distribution is close, conditional distribution of the two domains are also close, i.e., $P(\mathbf{Y}_s|\phi(\mathbf{X}_s)) \approx P(\mathbf{Y}_t|\phi(\mathbf{X}_t))$. Based on the above strict assumptions, TCA employs MMD to learn a unified transformation ϕ, which projects data into a low-dimensional subspace for reducing marginal distribution shift.

In conditional distribution adaptation, conditional distribution of data is different, i.e., $P(\mathbf{Y}_s|\phi(\mathbf{X}_s)) \neq P(\mathbf{Y}_t|\phi(\mathbf{X}_t))$, that is, marginal distribution of data is similar, i.e., $P(\phi(\mathbf{X}_s)) \approx P(\phi(\mathbf{X}_t))$, but there is a certain shift in each class. Stratified transfer learning (STL) [8] was proposed based on the idea of intraclass transfer. Most of the methods just learn a global feature transformation while ignoring similarity within each class. The method firstly uses majority voting to generate pseudo-labels for unlabeled target data; and then, intraclass correlation is used to adaptively reduce the dimension in RKHS, increasing the correlation between data in different domains; finally, through secondary labeling, accurate prediction of TD data is achieved. STL calculates MMD distance of each class in order to achieve intraclass transfer

$$C(\mathbf{X}_s, \mathbf{X}_t) = \sum_{l=1}^{c} \left\| \frac{1}{n_s^l} \sum_{\mathbf{x}_i \in \mathbf{X}_s^l} \phi(\mathbf{x}_i) - \frac{1}{n_t^l} \sum_{\mathbf{x}_j \in \mathbf{X}_t^l} \phi(\mathbf{x}_j) \right\|_{\mathrm{H}}^2 \tag{4}$$

where $\mathbf{X}_s^l$ and $\mathbf{X}_t^l$ are samples of class $l \in \{1, 2, \ldots, c\}$ in SD and TD, respectively, n_s^l and n_t^l are their corresponding number

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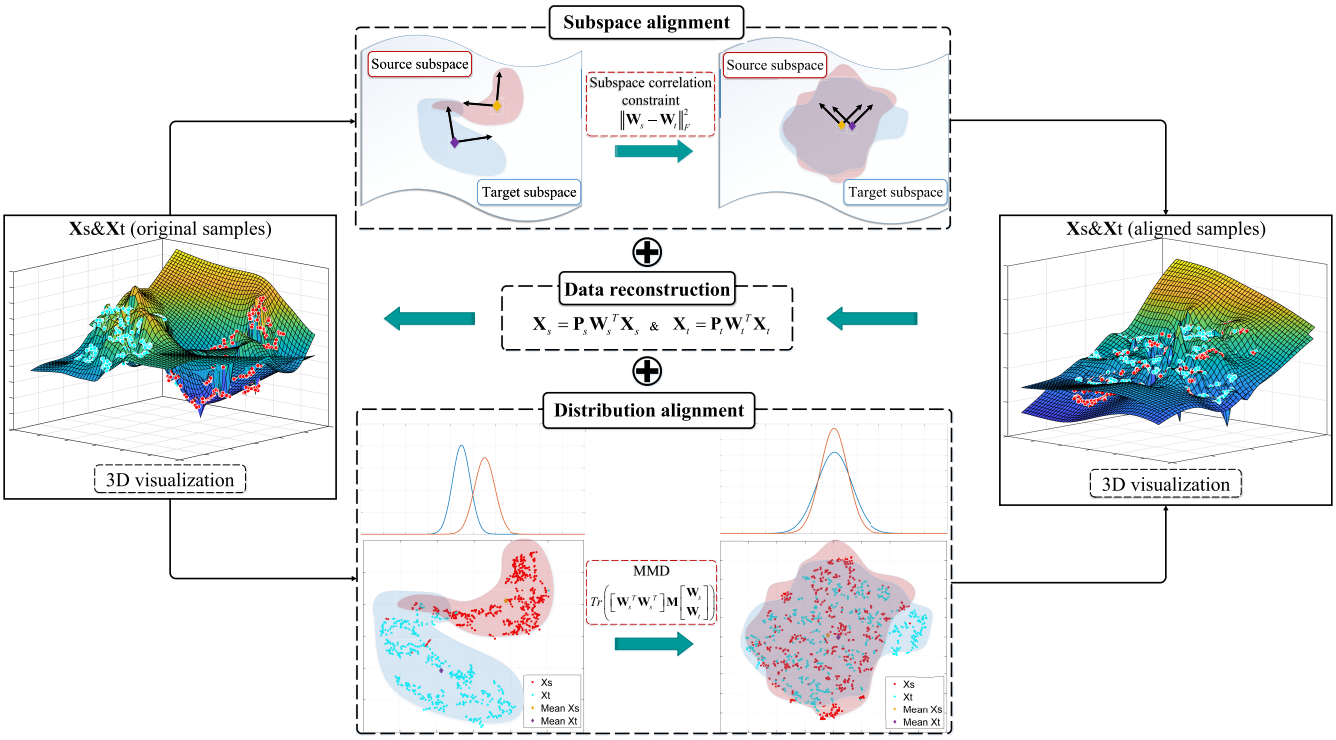


Fig. 2. Flowchart of the proposed DCA, including SA, distribution alignment and data reconstruction.

of samples from the lth class, and $\phi(\cdot)$ is a learned transformation to project the original samples to RKHS. By applying kernel tricks, (4) is rewritten as

$$\min_{\mathbf{W}} \sum_{c=1}^{C} \mathrm{tr}(\mathbf{W}^\top \mathbf{K}\mathbf{L}_c\mathbf{K}\mathbf{W}) + \lambda \mathrm{tr}(\mathbf{W}^\top\mathbf{W})$$
$$\text{s.t. } \mathbf{W}^\top\mathbf{K}\mathbf{H}\mathbf{K}\mathbf{W} = \mathbf{I} \tag{5}$$

where the first term denotes intraclass distance between SD and TD, $\mathbf{K}$ is the kernel gram matrix, $\mathbf{H}$ is a centering matrix and $\mathbf{L}_c$ is an intraclass matrix, and $\mathbf{I}$ is an identity matrix of appropriate size. The projection matrix can be constructed by solving a generalized eigen-decomposition problem.

C. Maximum Mean Discrepancy

MMD, as an effective nonparametric distance metric, is the most frequently used metric in DA [29]. It is a kernel learning method and measures the distance between two domains in RKHS. The MMD distance of two different domains is described as

$$M(\mathbf{X}_s, \mathbf{X}_t) = \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi(\mathbf{x}_i) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi(\mathbf{x}_j) \right\|_{\mathrm{H}}^2. \tag{6}$$

The physical meaning of MMD is to find the distance of mean values between SD and TD in RKHS. The Hilbert space is complete for the inner product of the function. After expanding the square, the inner product in RKHS can be converted into a kernel function, so MMD can be performed directly through the kernel function. There are some literature on improving typical MMD. For example, based on a single kernel transformation, Gretton *et al.* [30] proposed multiple-kernel MMD (MK-MMD), which used a linear combination of multiple kernels to obtain the optimal kernel. In addition, central moment discrepancy (CMD) [31] extended MMD to multiple orders, which utilized the difference of order moments to match the high-order central moments of the probability distribution, and eliminated the calculation of distance and kernel matrix.

III. Proposed DCA Classification Framework

The flowchart of the proposed DCA for cross-scene HSI classification is illustrated in Fig. 2, where red and blue dots represent data in SD and TD, respectively. SA and distribution alignment are performed cooperatively in the forward direction to reduce spectral shift, and data reconstruction is performed to preserve discriminative information in the subspace. Finally, more discriminative aligned samples are learned with simultaneously aligned the statistical and geometric features.

A. Algorithm Formulation

The main idea of a distribution alignment-based method is to seek a unified transform to project features of SD and TD to a domain invariant subspace, in order to reduce the distributional divergence of data in two domains, while retaining the original data properties. This type of method only uses the shared features; however, when SD and TD have only very few shared features, it is difficult to find an ideal domain invariant subspace. In the proposed DCA, projections of the two domains are learned, respectively. The respective projections are designed to reduce the spectral shift even if there is a quite difference between the two domains

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and enhance the adaptability of spectral shift. The objective function can be described as

$$\min_{\mathbf{W}_s,\mathbf{W}_t} \left\|\mathbf{W}_s^T\mathbf{X}_s - \mathbf{W}_t^T\mathbf{X}_t\right\|_F^2 + \lambda_1\|\mathbf{W}_s - \mathbf{W}_t\|_F^2 \tag{7}$$

where λ_1 is a regularization parameter controls the SA, $\mathbf{W}_s \in {}^{d\times k}$ and $\mathbf{W}_t \in {}^{d\times k}$ are the source and target subspaces (k is the dimension of subspace), and $\|\mathbf{W}_s - \mathbf{W}_t\|_F^2$ is a subspace correlation constraint. As mentioned, SA needs to learn a mapping relationship $\mathbf{M}$ between an SD subspace and a TD subspace. However, in the proposed DCA, this constraint is used to optimize both $\mathbf{W}_s$ and $\mathbf{W}_t$ instead of learning the transformation $\mathbf{M}$. Therefore, it is introduced to align the two subspaces, and the global geometric structure of two domains can be preserved; that is, subspace correlation, so that projection of SD and TD is not too far off.

Discriminant information in the multiband of two domains plays an important role in cross-scene HSI classification. In order to ensure the aligned samples in the low-dimensional subspace without loss of discriminative information, a data reconstruction item is further introduced in (7), which is rewritten as

$$\begin{aligned}\min_{\mathbf{W}_s,\mathbf{W}_t,\mathbf{P}_s,\mathbf{P}_t} & \left\|\mathbf{X}_s - \mathbf{P}_s\mathbf{W}_s{}^T\mathbf{X}_s\right\|_F^2 + \left\|\mathbf{X}_t - \mathbf{P}_t\mathbf{W}_t{}^T\mathbf{X}_t\right\|_F^2 \\ & + \lambda_1\|\mathbf{W}_s - \mathbf{W}_t\|_F^2 \\ \text{s.t. } & \mathbf{P}_s^T\mathbf{P}_s = \mathbf{I}, \quad \mathbf{P}_t^T\mathbf{P}_t = \mathbf{I}\end{aligned} \tag{8}$$

where $\mathbf{P}_s \in \mathbb{R}^{d\times k}$ and $\mathbf{P}_t \in \mathbb{R}^{d\times k}$ are two orthogonal reconstruction matrices. Learning the respective reconstruction matrices of the two domains can have more freedom to reconstruct the data of the corresponding domains. Compared with a unified reconstruction matrix, the reconstruction can be completed well even if the spectral shift of the SD and TD is large.

Equation (8) is called discriminant SA (DSA), which reduces the geometrical spectral shift, while ignoring the statistical distribution shift. Therefore, MMD is employed to realize cooperative alignment of subspace and distribution. The first part is the marginal distribution shift, i.e., interdomain MMD, which can be stated as

$$M(\mathbf{X}_s,\mathbf{X}_t) = \left\|\frac{1}{n_s}\sum_{i=1}^{n_s}\mathbf{W}_s^T\mathbf{x}_i - \frac{1}{n_t}\sum_{j=1}^{n_t}\mathbf{W}_t^T\mathbf{x}_j\right\|_F^2. \tag{9}$$

Interdomain MMD calculates difference in the mean of projected samples in k-dimensional subspace. The second part is conditional distribution shift, i.e., intraclass MMD, which can be described as

$$C(\mathbf{X}_s,\mathbf{X}_t) = \sum_{l=1}^{c}\left\|\frac{1}{n_s^l}\sum_{i=1}^{n_s^l}\mathbf{W}_s^T\mathbf{x}_i - \frac{1}{n_t^l}\sum_{j=1}^{n_t^l}\mathbf{W}_t^T\mathbf{x}_j\right\|_F^2 \tag{10}$$

where the target data $\mathbf{X}_t$ has no label, and target pseudo-labels are predicted by *K nearest neighbor* (KNN) classifier trained using the source data. After merging the conditional and marginal distribution shift, the distribution shift optimization item is obtained

$$\min_{\mathbf{W}} \ \mathrm{Tr}(\mathbf{W}^T\mathbf{X}\mathbf{M}\mathbf{X}^T\mathbf{W}) \tag{11}$$

where $\mathbf{W}$, $\mathbf{X}$ and $\mathbf{M}$ are denoted as $\begin{bmatrix}\mathbf{W}_s\\ \mathbf{W}_t\end{bmatrix}$, $[\mathbf{X}_s\mathbf{X}_t]$, and $\mathbf{M} = \begin{bmatrix}\mathbf{M}_s & \mathbf{M}_{st}\\ \mathbf{M}_{ts} & \mathbf{M}_t\end{bmatrix}$ respectively. The elements in $\mathbf{M}$ are denoted as

$$\mathbf{M}_s = \mathbf{L}_s + \sum_{l=1}^{c}\mathbf{L}_s^l, \quad \mathbf{L}_s = \frac{1}{n_s^2}\mathbf{I}_s\mathbf{I}_s^T$$

$$\left(\mathbf{L}_s^l\right)_{ij} = \begin{cases}\dfrac{1}{\left(n_s^l\right)^2}, & \mathbf{x}_i,\mathbf{x}_j \in \mathbf{X}_s^l\\ 0, & \text{otherwise}\end{cases} \tag{12}$$

$$\mathbf{M}_t = \mathbf{L}_t + \sum_{l=1}^{c}\mathbf{L}_t^l, \quad \mathbf{L}_t = \frac{1}{n_t^2}\mathbf{I}_t\mathbf{I}_t^T$$

$$\left(\mathbf{L}_t^l\right)_{ij} = \begin{cases}\dfrac{1}{\left(n_t^l\right)^2}, & \mathbf{x}_i,\mathbf{x}_j \in \mathbf{X}_t^l\\ 0, & \text{otherwise}\end{cases} \tag{13}$$

$$\mathbf{M}_{st} = \mathbf{L}_{st} + \sum_{l=1}^{c}\mathbf{L}_{st}^l, \quad \mathbf{L}_{st} = \frac{1}{n_s n_t}\mathbf{I}_s\mathbf{I}_t^T$$

$$\left(\mathbf{L}_{st}^l\right)_{ij} = \begin{cases}-\dfrac{1}{n_s^l n_t^l}, & \mathbf{x}_i \in \mathbf{X}_s^l, \mathbf{x}_j \in \mathbf{X}_t^l\\ 0, & \text{otherwise}\end{cases} \tag{14}$$

$$\mathbf{M}_{ts} = \mathbf{L}_{ts} + \sum_{l=1}^{c}\mathbf{L}_{ts}^l, \quad \mathbf{L}_{ts} = \frac{1}{n_s n_t}\mathbf{I}_t\mathbf{I}_s^T$$

$$\left(\mathbf{L}_{ts}^l\right)_{ij} = \begin{cases}-\dfrac{1}{n_s^l n_t^l}, & \mathbf{x}_i \in \mathbf{X}_t^l, \mathbf{x}_j \in \mathbf{X}_s^l\\ 0, & \text{otherwise}\end{cases}. \tag{15}$$

Substitute (11) into (8), and the objective function of the proposed DCA can be replaced as

$$\begin{aligned}\min_{\mathbf{W}_s,\mathbf{W}_t,\mathbf{P}_s,\mathbf{P}_t} & \left\|\mathbf{X}_s - \mathbf{P}_s\mathbf{W}_s^T\mathbf{X}_s\right\|_F^2 + \left\|\mathbf{X}_t - \mathbf{P}_t\mathbf{W}_t^T\mathbf{X}_t\right\|_F^2 \\ & + \lambda_1\|\mathbf{W}_s - \mathbf{W}_t\|_F^2 + \lambda_2\mathrm{Tr}(\mathbf{W}^T\mathbf{X}\mathbf{M}\mathbf{X}^T\mathbf{W}) \\ \text{s.t. } & \mathbf{P}_s^T\mathbf{P}_s = \mathbf{I}, \quad \mathbf{P}_t^T\mathbf{P}_t = \mathbf{I}\end{aligned} \tag{16}$$

where λ_2 is a regularization parameter that adjusts the contribution of distribution alignment. After iteratively optimizing variables and learning $\mathbf{W}_s$ and $\mathbf{W}_t$, KNN classifier trained using aligned source samples is employed to classify the aligned target data.

B. Algorithm Optimization

Alternating direction method of multipliers (ADMM) strategy [32] is used to iteratively optimize (16) to minimize the loss of one variable when fixing other variables. There are four variables that need to be optimized, i.e., $\mathbf{W}_s$, $\mathbf{W}_t$, $\mathbf{P}_s$, and $\mathbf{P}_t$. Details of the optimization steps are stated as follows.

1) Updating $\mathbf{W}_s$*:* Updating $\mathbf{W}_s$ by solving the following problem when fixing other variables:

$$\begin{aligned}\min_{\mathbf{W}_s} & \left\|\mathbf{X}_s - \mathbf{P}_s\mathbf{W}_s^T\mathbf{X}_s\right\|_F^2 + \lambda_1\|\mathbf{W}_s - \mathbf{W}_t\|_F^2 \\ & + \lambda_2 Tr\left(\mathbf{W}^T\mathbf{X}\begin{bmatrix}\mathbf{M}_s & \mathbf{M}_{st}\\ \mathbf{M}_{ts} & \mathbf{0}\end{bmatrix}\mathbf{X}^T\mathbf{W}\right) \\ \text{s.t. } & \mathbf{P}_s^T\mathbf{P}_s = \mathbf{I}, \quad \mathbf{P}_t^T\mathbf{P}_t = \mathbf{I}.\end{aligned} \tag{17}$$

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Equation (17) has a closed-form solution, which can be found by setting the first partial derivative of $\mathbf{W}_s$ and making it equal to zero. The expression of $\mathbf{W}_s$ is written as

$$\begin{aligned}\mathbf{W}_s &= \left[\mathbf{X}_s\mathbf{X}_s^T + \lambda_1\mathbf{I} + \lambda_2\mathbf{G}_1\right]^{-1} \\ &\quad \times \left(\mathbf{X}_s\mathbf{X}_s^T\mathbf{P}_s + \lambda_1\mathbf{W}_t - \lambda_2\mathbf{G}_2\mathbf{W}_t\right) \\ \mathbf{G}_1 &= \mathbf{X}_s\mathbf{M}_s\mathbf{X}_s^T + \mathbf{X}_s^T\mathbf{M}_s^T\mathbf{X}_s \\ \mathbf{G}_2 &= \mathbf{X}_s\mathbf{M}_{st}\mathbf{X}_t^T + \mathbf{X}_s\mathbf{M}_{ts}^T\mathbf{X}_t^T. \end{aligned} \tag{18}$$

2) Updating $\mathbf{W}_t$*:* Similarly, the optimization problem of $\mathbf{W}_s$ is written as

$$\begin{aligned}\min_{\mathbf{W}_t}\ & \left\|\mathbf{X}_t - \mathbf{P}_t\mathbf{W}_t^T\mathbf{X}_t\right\|_F^2 + \lambda_1\|\mathbf{W}_s - \mathbf{W}_t\|_F^2 \\ &+ \lambda_2\mathrm{Tr}\left(\mathbf{W}^T\mathbf{X}\begin{bmatrix}\mathbf{0} & \mathbf{M}_{st}\\ \mathbf{M}_{ts} & \mathbf{M}_t\end{bmatrix}\mathbf{X}^T\mathbf{W}\right) \\ \text{s.t. } & \mathbf{P}_s^T\mathbf{P}_s = \mathbf{I}, \quad \mathbf{P}_t^T\mathbf{P}_t = \mathbf{I}. \end{aligned} \tag{19}$$

The solving process of $\mathbf{W}_t$ is similar to that of $\mathbf{W}_s$, and its expression is stated as

$$\begin{aligned}\mathbf{W}_t &= \left[\mathbf{X}_t\mathbf{X}_t^T + \lambda_1\mathbf{I} + \lambda_2\mathbf{G}_3\right]^{-1} \\ &\quad \times \left(\mathbf{X}_t\mathbf{X}_t^T\mathbf{P}_t + \lambda_1\mathbf{W}_s - \lambda_2\mathbf{G}_4\mathbf{W}_s\right) \\ \mathbf{G}_3 &= \mathbf{X}_t\mathbf{M}_t\mathbf{X}_t^T + \mathbf{X}_t^T\mathbf{M}_t^T\mathbf{X}_t \\ \mathbf{G}_4 &= \mathbf{X}_t\mathbf{M}_{st}^T\mathbf{X}_s^T + \mathbf{X}_t\mathbf{M}_{ts}\mathbf{X}_s^T. \end{aligned} \tag{20}$$

3) Updating $\mathbf{P}_s$*:* Fixing $\mathbf{W}_s$, $\mathbf{W}_t$ and $\mathbf{P}_t$, the optimization of $\mathbf{P}_s$ is described as

$$\min_{\mathbf{P}_s} \left\|\mathbf{X}_s - \mathbf{P}_s\mathbf{W}_s^T\mathbf{X}_s\right\|_F^2 \quad \text{s.t. } \mathbf{P}_s^T\mathbf{P}_s = \mathbf{I}. \tag{21}$$

Equation (21) can be solved by singular value decomposition (SVD), $\mathrm{SVD}(\mathbf{X}_s\mathbf{X}_s^T\mathbf{W}_s) = \mathbf{U}\mathbf{S}\mathbf{V}^T$, the solution of $\mathbf{P}_s$ is $\mathbf{P}_s = \mathbf{U}\mathbf{V}^T$.

4) Updating $\mathbf{P}_t$*:* Similarly, the optimization of $\mathbf{P}_t$ is written as

$$\min_{\mathbf{P}_t} \left\|\mathbf{X}_t - \mathbf{P}_t\mathbf{W}_t^T\mathbf{X}_t\right\|_F^2 \quad \text{s.t. } \mathbf{P}_t^T\mathbf{P}_t = \mathbf{I}. \tag{22}$$

SVD is also employed to solve (22), and the solution of $\mathbf{P}_t$ is $\mathbf{P}_t = \mathbf{U}\mathbf{V}^T$.

C. Analysis on Proposed DCA

The spectral shift is the most critical issue in cross-scene HSI classification task; that is, pixels from the same class in SD and TD have different spectral signatures. Fig. 3(a), (d), (g), (j), and (m) illustrate such phenomena, which are two-dimensional visualization of the original samples obtained from the HyRANK and Houston data, red and blue dots represent the source and target data, respectively, and purple and yellow dots are centroids of the distribution. Obviously, distributions of the original samples from the same class in the two domains are quite different, and their centroids are far apart.

To demonstrate the alignment performance, Fig. 3 illustrates the distribution of samples aligned by DSA and the proposed DCA. They have mitigated the spectral shift problem. In contrast, the alignment performance of the proposed DCA is obviously better, which indicates that the proposed cooperative alignment of subspace and distribution is more effective than SA only. In addition, it can be seen that the overall distribution of the same class corresponding to the two domains of the Houston data in (l) and (o) does not match well, because the intra-class MMD does not work in the case that two MMD constraints have the equal weight.

Algorithm 1 Proposed DCA Classification Framework

Input: Source data $\mathbf{X}_s$, target data $\mathbf{X}_t$, source label $\mathbf{L}_s$, and regularization parameters λ_1, λ_2.
Initialization: Matrix $\mathbf{W}_s$, $\mathbf{W}_t$ with random values.
Step 1: Initialize target pseudo-labels $\mathbf{L}_t$ by KNN trained on original source data
while not converged **or** max iter **do**
Step 2: Construct $\mathbf{M}_s$, $\mathbf{M}_t$, $\mathbf{M}_{st}$ and $\mathbf{M}_{ts}$ according to (12), (13), (14) and (15)
Step 3: Update the reconstruction matrix $\mathbf{P}_s$ and $\mathbf{P}_t$ by solving (21) and (22)
Step 4: Update the projection matrix $\mathbf{W}_s$ and $\mathbf{W}_t$ according to (18) and (20)
Step 5: Update target pseudo-labels $\mathbf{L}_t$ by KNN trained on $\mathbf{W}_s^T\mathbf{X}_s$
end while
Output: $\mathbf{W}_s$, $\mathbf{W}_t$ and $\mathbf{L}_t$.

TABLE I
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HOUSTON DATA SET

Class		Number of Samples	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, three cross-scene HSI data sets, i.e., the Houston data, HyRANK data, and Shanghai-Hangzhou data, are conducted to verify the effectiveness of the proposed DCA. Several classic and state-of-the-art unsupervised domain adaptive algorithms are employed for comparison algorithms, including KNN, TCA [16], CORAL [13], STL [8], scatter component analysis (SCA) [33], joint distribution adaptation (JDA) [34], joint geometrical and statistical alignment (JGSA) [35], and manifold embedded distribution alignment (MEDA) [36]. The class-specific accuracy (CA), the overall accuracy (OA), and the Kappa coefficient (KC) are employed to evaluate the classification performance.

A. Experiment Data

1) Houston: The data set includes Houston 2013 [37] and Houston 2018 [38] scenes, which were obtained by different sensors on the University of Houston campus and its vicinity

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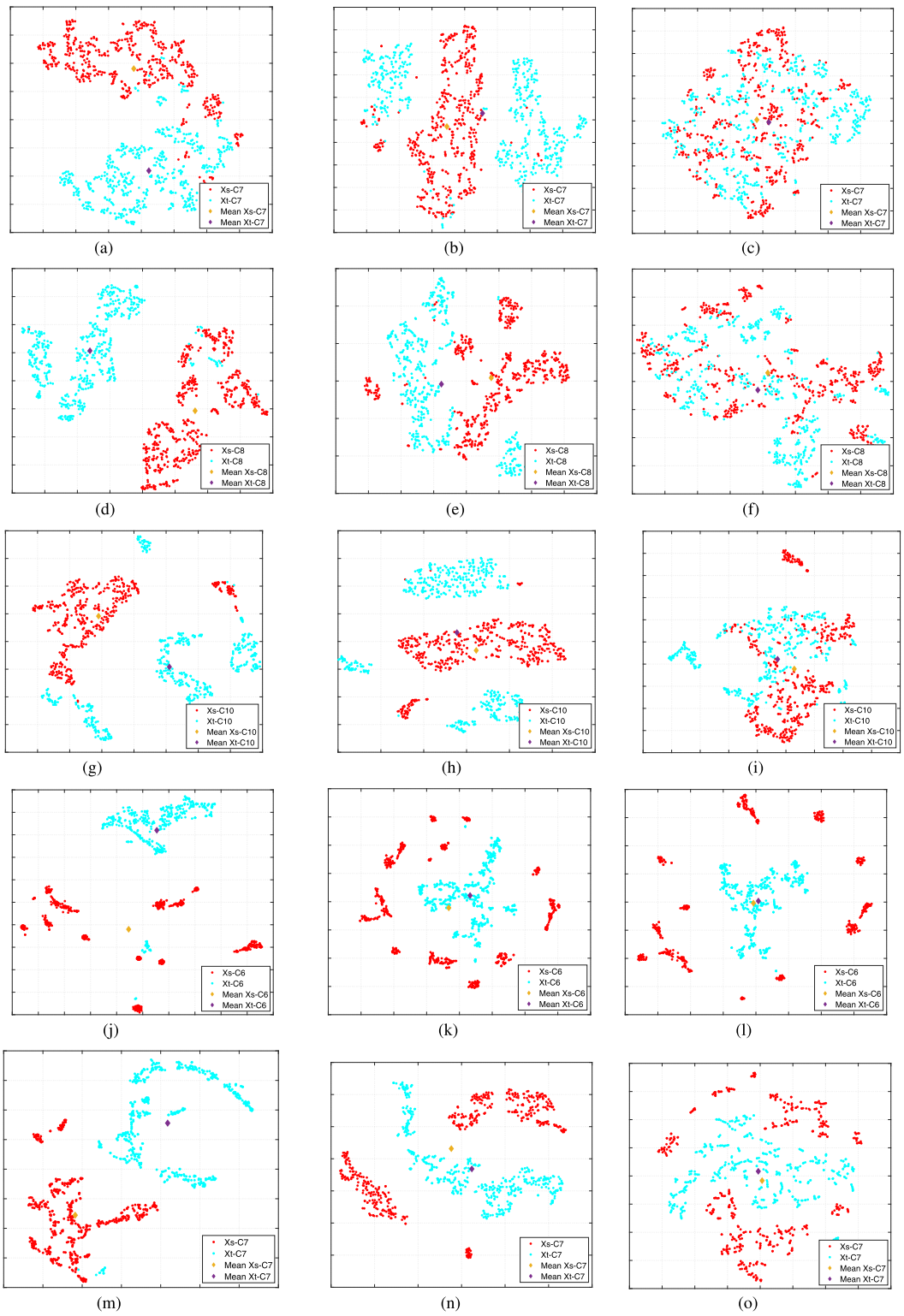


Fig. 3. Alignment performance of the proposed DCA using the HyRANK and Houston data. (a) Original samples from HyRANK (Class 7). (b) Aligned samples by DSA (Class 7). (c) Aligned samples by DCA (Class 7). (d) Original samples from HyRANK (Class 8). (e) Aligned samples by DSA (Class 8). (f) Aligned samples by DCA (Class 8). (g) Original samples from HyRANK (Class 10). (h) Aligned samples by DSA (Class 10). (i) Aligned samples by DCA (Class 10). (j) Original samples from Houston (Class 6). (k) Aligned samples by DSA (Class 6). (l) Aligned samples by DCA (Class 6). (m) Original samples from Houston (Class 7). (n) Aligned samples by DSA (Class 7). (o) Aligned samples by DCA (Class 7).

in different years. The Houston 2013 data set is composed of 349×1905 pixels, including 144 spectral bands, the wavelength range is 380–1050 nm, and the image spatial resolution is 2.5 m. The Houston 2018 data set has the same wavelength range but contains 48 spectral bands, and the image has a spatial resolution of 1 m. There are seven consistent classes in their scene. We extract 48 spectral bands from Houston 2013 scene corresponding to Houston 2018 scene, and select

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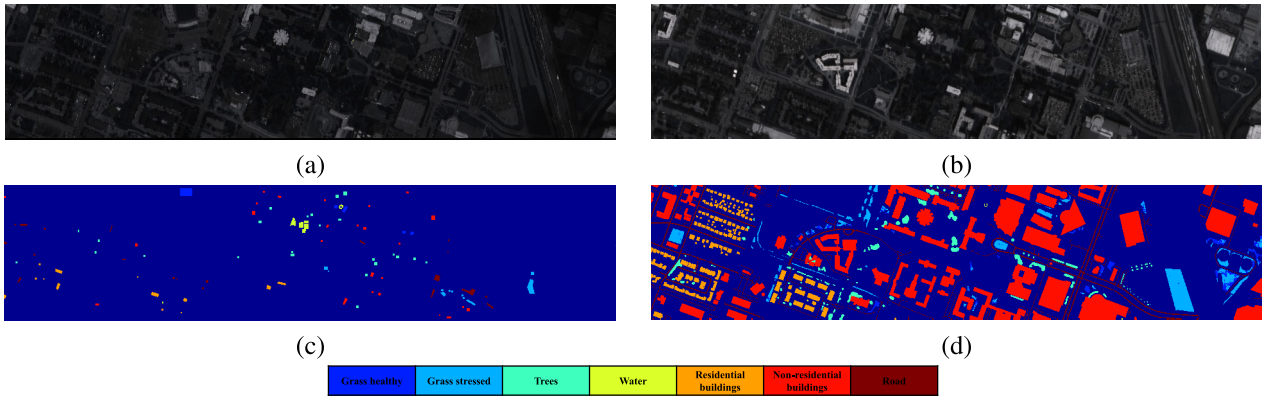


Fig. 4. False-color image and ground truth map of Houston. (a) False-color image of Houston 2013. (b) False-color image of Houston 2018. (c) Ground truth map of Houston 2013. (d) Ground truth map of Houston 2018.

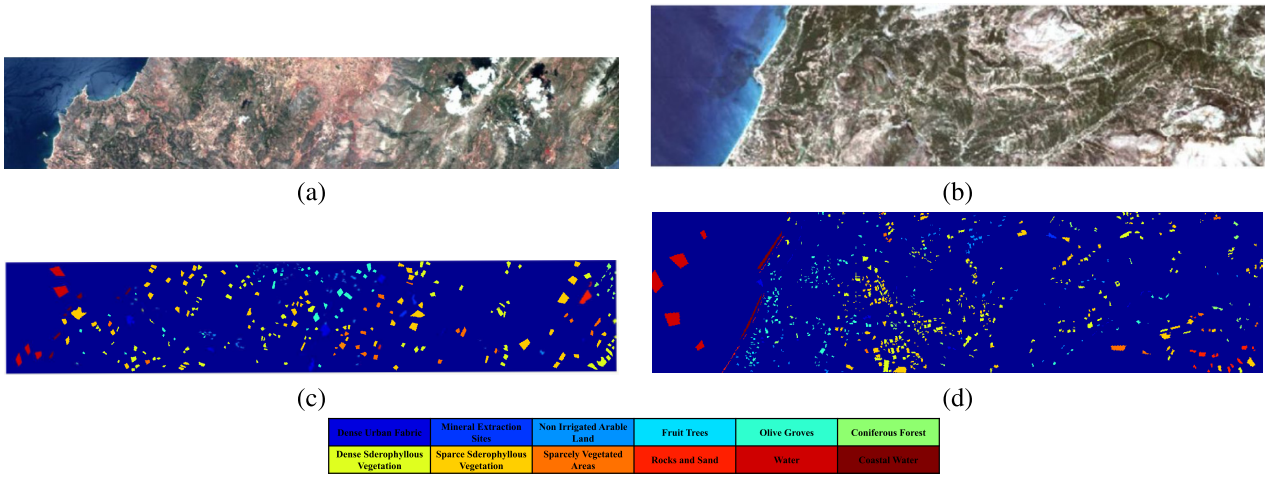


Fig. 5. False-color image and ground truth map of HyRANK. (a) False-color image of Dioni. (b) False-color image of Loukia. (c) Ground truth map of Dioni. (d) Ground truth map of Loukia.

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HYRANK DATA SET

Class		Number of Samples	
No.	Name	Dioni (Source)	Loukia (Target)
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sderophyllous Vegetation	5035	2996
8	Sparce Sderophyllous Vegetation	6374	2361
9	Sparcely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
Total		20024	10317

TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE SHANGHAI-HANGZHOU DATA SET

Class		Number of Samples	
No.	Name	Shanghai (Source)	Hangzhou (Target)
1	Water	123123	18043
2	Land/Building	161689	77450
3	Plant	83188	40207
Total		368000	135700

the overlapping area of 209 × 955. The name of land cover classes and the number of samples are listed in Table I. Additionally, their false-color representations and ground truth maps are shown in Fig. 4.

2) HyRANK: The HyRANK data set has been developed in the framework of the International Society for Photogrammetry and RS (ISPRS) Scientific Initiatives [39]. The satellite HSI collected by the Hyperion sensor (EO-1, USGS) has 176 spectral bands. The two labeled scenes are Dioni and Loukia, which are composed of 250 × 1376 pixels and 249 × 945 pixels, respectively. There are twelve consistent classes, which are listed in Table II. The vegetation classes are prone to be misclassified. The false-color representations and ground truth maps of two scenes are shown in Fig. 5.

3) Shanghai-Hangzhou: The Shanghai and Hangzhou data sets were captured by the EO-1 Hyperion hyperspectral sensor [40], which retains 198 bands after removing the bad bands. The size of the Shanghai data image is 1600 × 230, which includes roads, buildings, plants, and the waters of the Yangtze River and Huangpu River. The Hangzhou data image size

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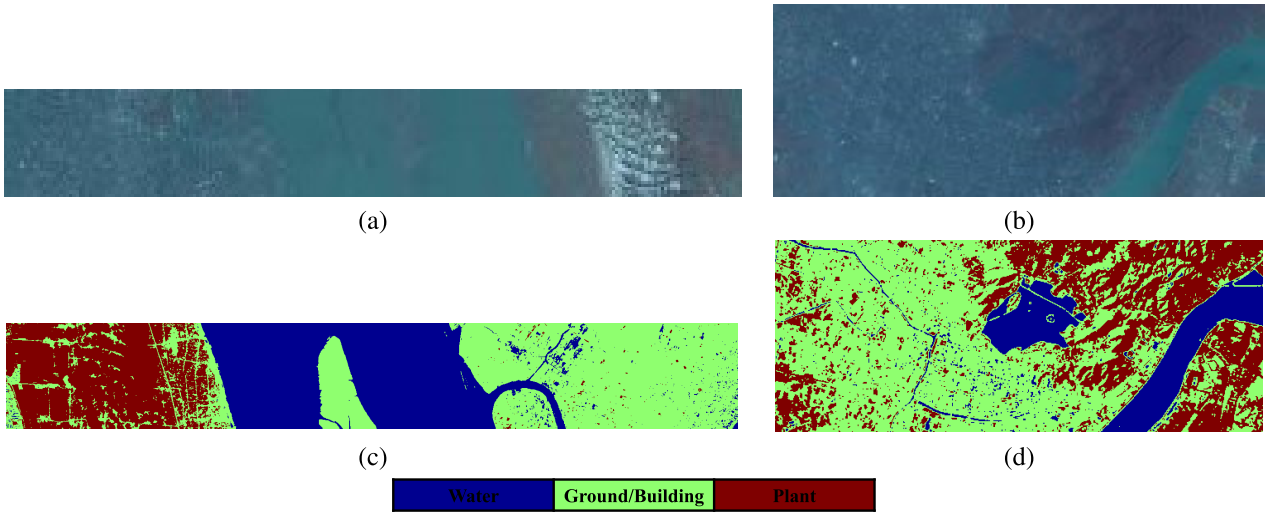


Fig. 6. False-color image and ground truth map of Shanghai-Hangzhou. (a) False-color image of Shanghai. (b) False-color image of Hangzhou. (c) Ground truth map of Shanghai. (d) Ground truth map of Hangzhou.

is 590×230, including roads, buildings, plants, West Lake, and Qiantang River basin. Three common land cover classes were selected for cross-scene classification, which are listed in Table III, and their false-color representations and ground truth maps are shown in Fig. 6.

B. Parameter Tuning

According to the proposed DCA, adjustable parameters are the regularization parameters, i.e., λ_1, λ_2, and the dimension of subspace, i.e., k. The same strategy as [34] is adopted; that is, empirical search method is considered to find the optimal hyperparameters. The value ranges of λ_1 and λ_2 are $\{1e-4, 1e{-}3, 1e{-}2, 1e{-}1, 1e{-}0, 5e{+}0, 1e{+}1, 1e{+}2, 5e{+}2\}$. Fig. 7 illustrates classification accuracy of the proposed DCA in three experimental data with varying values of parameters λ_1 and λ_2. In order to reduce the pressure of parameter optimization, we select suboptimal parameters. The λ_1 and λ_2 are parameters that control the SA and the distribution alignment, respectively. It can be seen from Fig. 7 that the proposed method with parameter λ_1 in the range of $1e-4$ to $1e-2$ can achieve suboptimal classification performance, which shows that DCA is less sensitive to λ_1. The λ_2 corresponds to three data sets with their own suboptimal ranges, that is, DCA is more sensitive to λ_2. The above analysis shows that the distribution alignment in the proposed DCA plays a more important role in reducing the spectral shift than the SA. Therefore, suboptimal parameters λ_1 and λ_2 corresponding to the Houston data are $1e{-}4$ and $1e{-}3$; for the HyRANK data, they are $1e{-}4$ and $1e+1$; for the Shanghai-Hangzhou data, they are $1e-4$ and $1e-2$, respectively.

Relationship between subspace dimension k and OA in the three real experimental data sets are shown in Fig. 8, which is obtained under fixed regularization parameters, i.e., λ_1 and λ_2. The horizontal axis in Fig. 8 represents the subspace dimension, from the smallest dimension to the largest dimension corresponding to each experimental data, and the vertical axis represents the classification accuracy. It is interesting to observe that the performance is pretty stable when the subspace dimension is around 20 or 30. Thus, the parameter is empirically set to 20.

C. Performance on Synthetic Data

The synthetic data set is firstly employed to demonstrate that the proposed DCA can reduce the domain shift while maintaining the global geometric structure. Each class of SD and TD is randomly generated by Gaussian distribution, i.e., $x_i^l \sim N(\mu^l, \sigma^l)$ (μ^l and σ^l are mean and standard deviation of the class l), and global mean between the two domains and mean of the 3rd class are shifted.

The synthetic data and classification results obtained with different methods are shown in Fig. 9. TCA and JDA, distributed alignment-based methods, reduce the domain shift of the 2nd class, while mixing the 1st class and the 3rd class together, probably because there is no domain invariant subspace to reduce the domain shift while preserving the original data properties. CORAL, an SA-based method, significantly reduces the domain shift of the 3rd class, but mixing the 3rd class with the 1st class. STL and SCA completely confuse the three classes. MEDA fails to reduce the domain shift of the 3rd class and cannot maintain the global geometry of the data. JGSA performs well in both statistical and geometric alignment, but when compared with the proposed DCA, the discriminative information within the class is slightly worse, e.g., intraclass compactness. After DSA aligns the subspaces, there is still a domain shift. Therefore, SA is not optimal for reducing domain shift without considering the distribution alignment. The proposed DCA performs SA and distribution alignment cooperatively under the constraint of data reconstruction, which can effectively reduce the difference in the two domains.

D. Performance on Real-World HSI Data

Furthermore, real-world HSI data sets are employed to evaluate the performance of the proposed DCA. We use nor-

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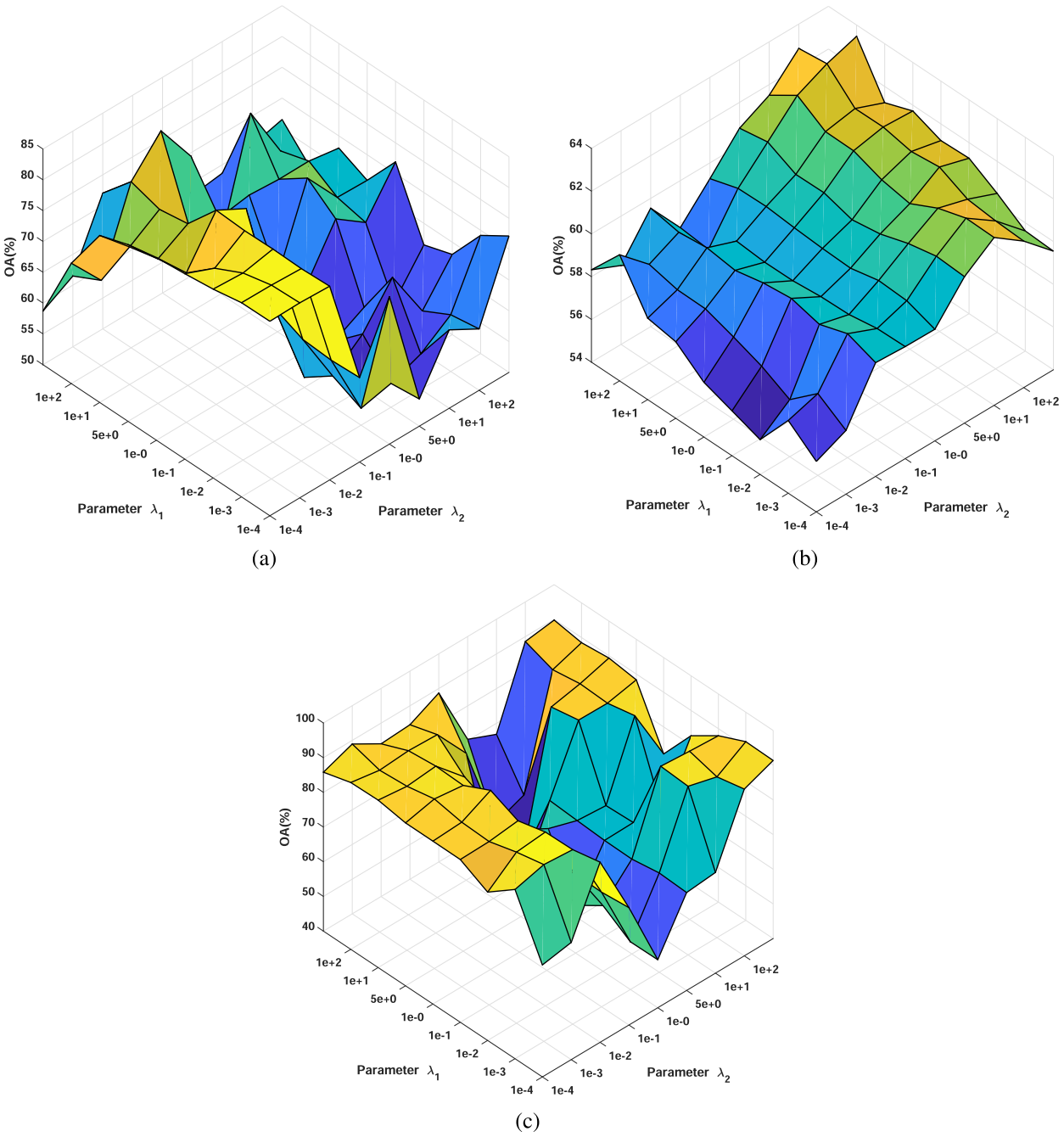


Fig. 7. Parameter tuning of λ_1 and λ_2 for the proposed DCA using all the three experimental data. (a) Houston. (b) HyRANK. (c) Shanghai-Hangzhou.

TABLE IV

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018

Class	Classification algorithms									
	KNN	TCA	CORAL	STL	SCA	JDA	JGSA	MEDA	DSA	DCA
1	99.04	80.49	78.86	88.84	0.00	79.75	0.00	100	76.42	82.79±2.37
2	42.23	94.23	94.93	94.41	2.56	94.60	16.51	1.29	96.19	99.98±0.02
3	22.52	67.21	66.70	56.04	88.11	66.92	91.65	5.93	58.93	54.72±0.60
4	100	100	100	100	100	100	95.45	100	100	100±0
5	55.34	63.85	64.28	66.30	0.00	63.89	21.81	79.33	89.06	85.00±0.13
6	81.27	67.06	68.14	73.19	89.41	67.59	85.55	69.12	73.67	76.13±0.56
7	23.05	62.20	61.21	64.46	0.00	61.15	8.03	75.60	83.75	97.47±1.00
OA (%)	65.51	69.01	69.60	72.92	59.41	69.22	61.67	62.20	77.76	**80.79±0.26**
KC (κ)	40.74	53.45	54.04	57.87	23.35	53.59	33.23	44.69	66.30	**71.00±0.31**

malized spectral data for training and testing in the experiment. Comparison algorithms include KNN, TCA, CORAL, STL, SCA, JDA, JGSA, and MEDA. In particular, CORAL has no parameters, TCA, STL, and JDA have one hyperparame-

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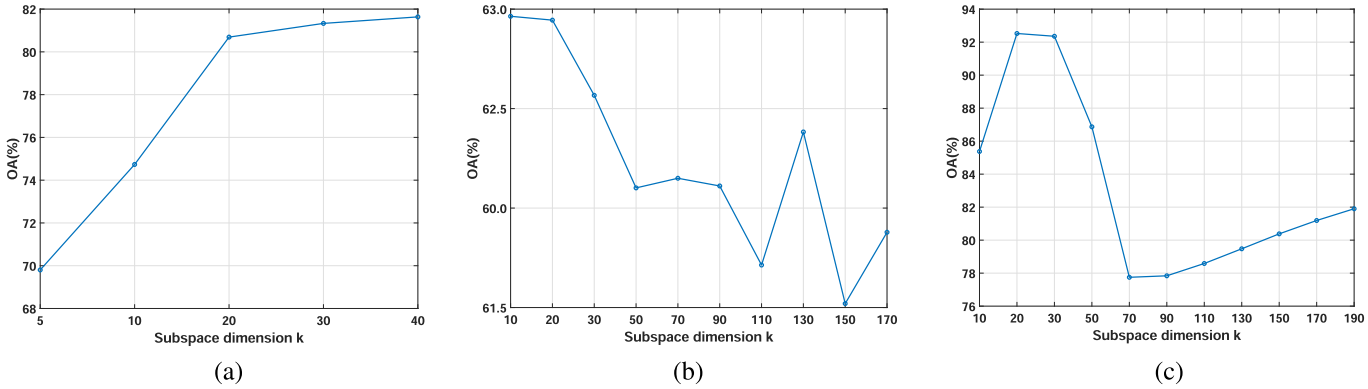


Fig. 8. Parameter tuning of the subspace dimension k for the proposed DCA using the three experimental data. (a) Houston. (b) HyRANK. (c) Shanghai-Hangzhou.

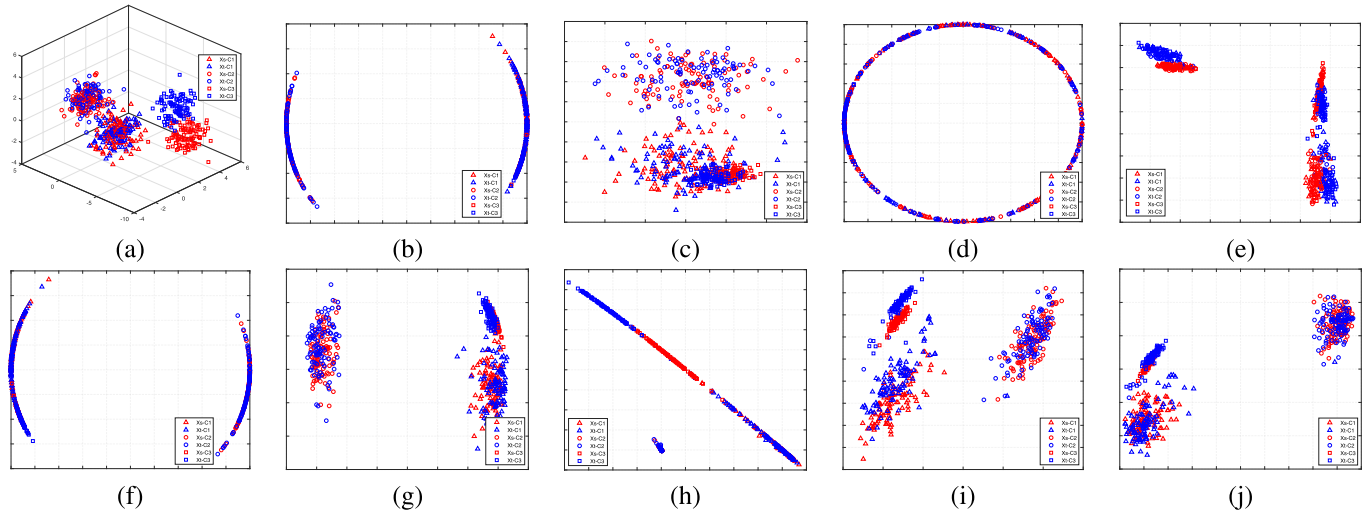


Fig. 9. Visualization and classification results for the synthetic data obtained with different methods including (a) synthetic data, (b) TCA (93.00%), (c) CORAL (95.33%), (d) STL (59.00%), (e) SCA (96.67%), (f) JDA (96.67%), (g) JGSA (97.53%), (h) MEDA (98.00%), (i) DSA (96.67%), and (j) proposed DCA (98.00%).

TABLE V

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE LOUKIA

Class	Classification algorithms									
	KNN	TCA	CORAL	STL	SCA	JDA	JGSA	MEDA	DSA	DCA
1	2.43	16.99	15.05	1.46	15.05	15.53	72.33	0.49	17.96	56.8±5.23
2	90.74	81.48	51.85	31.48	0.0	57.41	59.26	0.00	75.93	16.48±0.59
3	55.87	69.25	36.38	42.02	36.85	43.43	14.32	0.70	64.32	37.75±3.99
4	77.22	50.63	0.00	0.00	25.32	1.27	7.59	0.00	8.86	1.9±1.08
5	1.54	9.39	0.18	0.00	0.63	0.63	54.56	4.97	0.99	12.32±2.36
6	21.33	2.84	27.49	20.14	0.00	11.85	0.00	0.24	11.85	4.72±1.01
7	67.56	72.1	80.47	83.61	66.52	78.24	36.55	78.04	77.47	68.06±0.55
8	4.57	60.23	33.76	57.35	59.13	45.11	51.55	90.77	52.18	72.03±2.02
9	14.29	18.80	31.33	30.83	6.27	28.82	82.96	1.75	21.30	41.98±5.40
10	5.08	36.20	39.74	8.39	41.06	44.37	8.39	2.65	69.09	63.84±5.39
11	96.48	100	100	100	100	100	100	100	100	100±0
12	100	100	100	100	0	100	100	100	100	100±0
OA (%)	43.01	59.76	54.85	59.30	50.48	56.65	51.83	61.78	59.95	**62.57±0.49**
KC (κ)	35.41	51.33	45.26	49.67	40.75	47.16	43.05	50.82	51.28	**53.84±0.64**

ter, and JGSA and MEDA have two hyperparameters to be adjusted. In the following experiment, candidate parameter set is $\{1e-6,\ 1e-5,\ 1e-4,\ 1e-3,\ 1e-2,\ 1e-1,\ 1e-0,\ 5e+0,\ 1e+1,\ 1e+2,\ 5e+2\}$ and using the empirical search method to find the corresponding optimal parameters. SCA finds the optimal parameters according to the strategy of Hyper-Parameter Settings in [33]. The setting of the maximum number of iterations: JDA is set to 10, JGSA is set to 50, MEDA is set to 10, and DCA is set to 200. In addition, due to the large sample size of the three data sets, a large amount of memory is required to construct an $n_s \times n_s$ or $n_t \times n_t$ matrix when calculating MMD. To alleviate the memory problem, the data are randomly divided into ten subsets, and one subset is used for each iteration. Finally, the aligned sample is obtained, and the simple KNN is used for classification.

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TABLE VI

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HANGZHOU

Class	Classification algorithms									
	KNN	TCA	CORAL	STL	SCA	JDA	JGSA	MEDA	DSA	DCA
1	100	99.97	99.82	99.73	100	100	99.86	89.66	98.79	96.21±1.42
2	73.53	87.93	74.53	78.10	67.13	81.64	91.36	99.75	79.60	92.14±4.78
3	2.47	54.39	15.14	33.35	31.15	48.86	78.80	59.84	97.21	90.91±5.39
OA (%)	56.00	79.60	60.30	67.72	60.84	74.37	88.77	86.58	87.37	**91.83±1.84**
KC (κ)	20.80	63.63	29.52	43.25	34.16	55.25	80.37	74.71	78.91	**85.53±2.82**

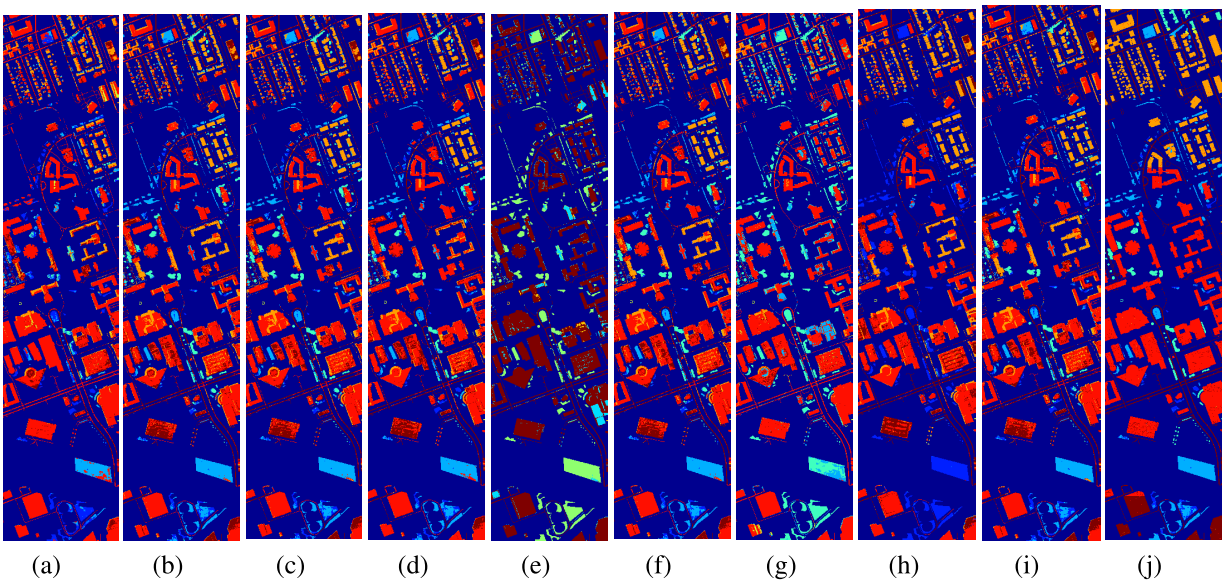


Fig. 10. Visualization and classification maps for the target scene Houston 2018 obtained with different methods including (a) KNN (65.51%), (b) TCA (69.01%), (c) CORAL (69.60%), (d) STL (72.92%), (e) SCA (59.41%), (f) JDA (69.22%), (g) JGSA (61.67%), (h) MEDA (62.20%), (i) DSA (76.54%), and (j) DCA (80.79%).

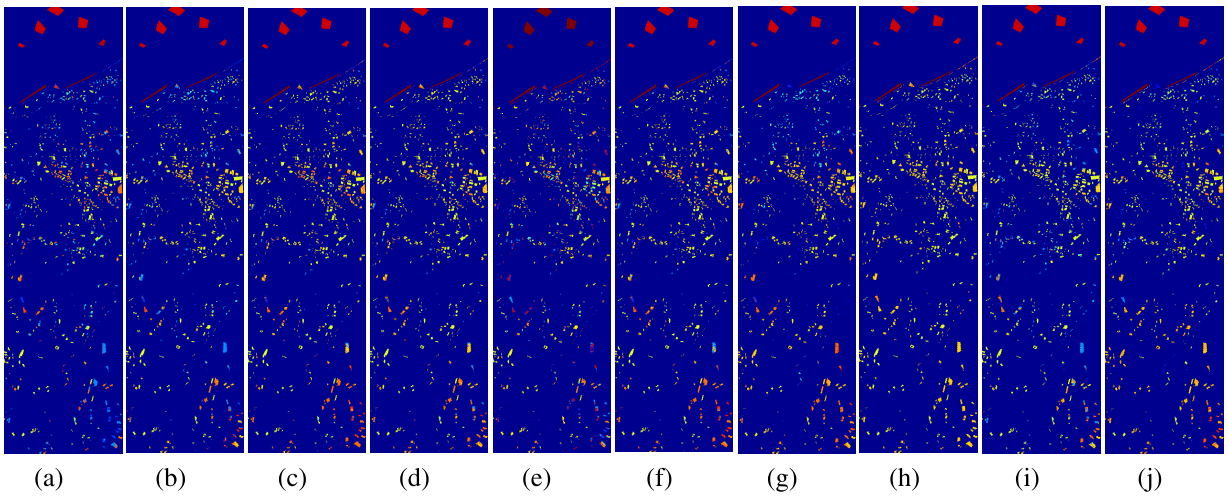


Fig. 11. Visualization and classification maps for the target scene Luokia obtained with different methods including (a) KNN (44.06%), (b) TCA (59.76%), (c) CORAL (54.85%), (d) STL (59.30%), (e) SCA (50.48%), (f) JDA (56.65%), (g) JGSA (51.83%), (h) MEDA (61.78%), (i) DSA (59.65%), and (j) DCA (62.37%).

Tables IV–VI provide class-specific classification performance of aforementioned approaches on the three target scenes, and report the average precision and standard deviation. DCA uses SD and TD data and SD labels during training. The SD training data are selected as follows: all the samples are used in the Houston 2013 and Dioni, and 180 samples are selected for each class in the Shanghai data. Obviously, the proposed DCA provides better classification performance when compared with other algorithms, especially in the Houston 2018 and Hangzhou scenes. It can be seen from Figs. 10–12 the obtained classification maps, that some areas have less noise and more accurate results, such as the 2nd class (Grass stressed), the 6th class (Nonresidential buildings), and the 7th class (Road) in the Houston 2018 data. Please note that compared with the other two data sets, the Shanghai-Hangzhou data set is very noisy, and the CA of DCA in Table VI are all

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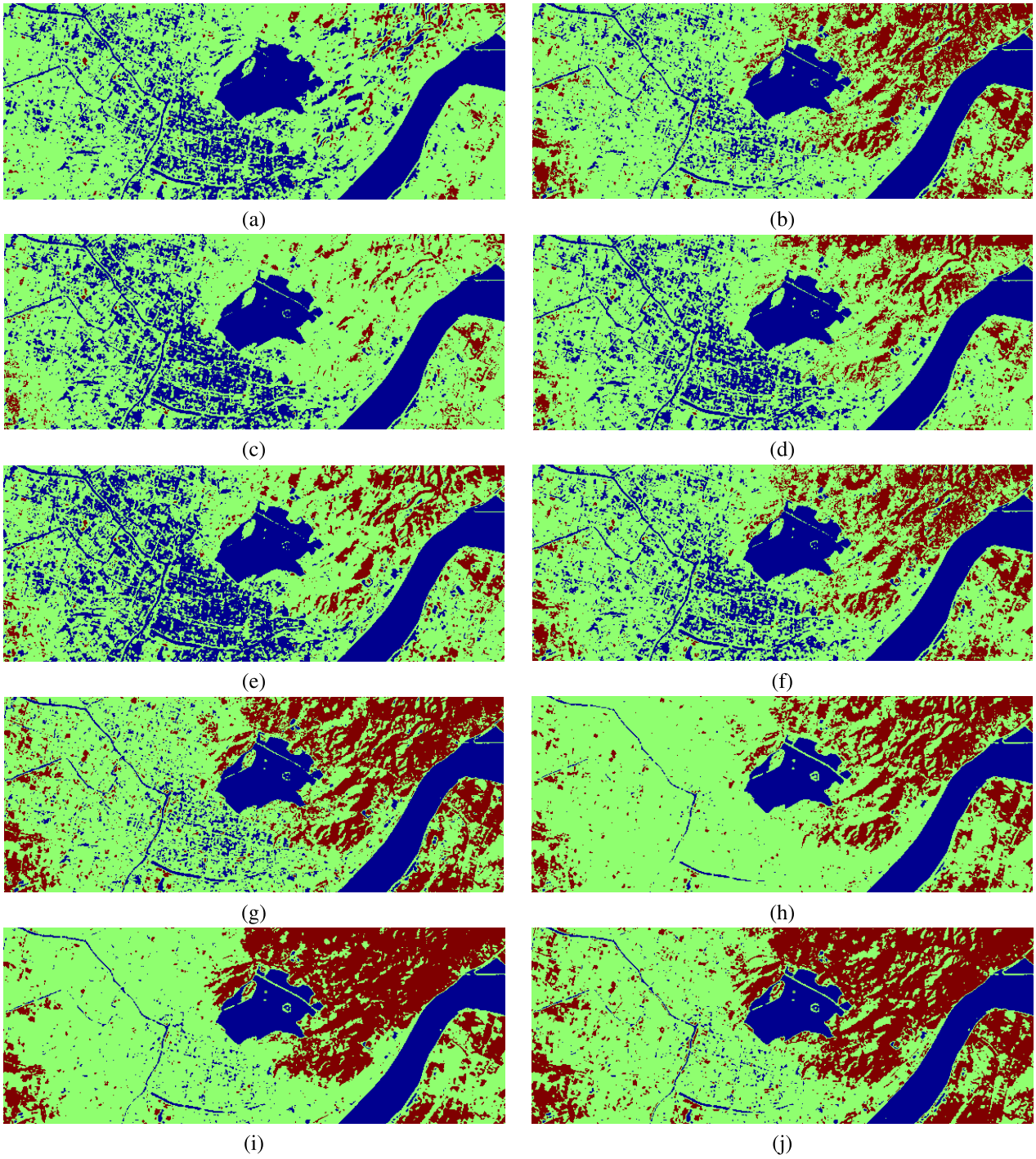


Fig. 12. Visualization and classification maps for the target scene Hangzhou obtained with different methods including (a) KNN (58.36%), (b) TCA (79.60%), (c) CORAL (60.30%), (d) STL (67.72%), (e) SCA (60.84%), (f) JDA (74.37%), (g) JGSA (88.77%), (h) MEDA (86.58%), (i) DSA (87.51%), and (j) DCA (92.38%).

around 90%, which also shows that DCA is more robust than other methods. In addition, the performance in the HyRANK data set is poor since some classes are similar, such as the 4th class (Fruit Trees), the 5th class (Olive Groves), and the 6th class (Coniferous Forest). Comparing DSA and the proposed DCA, it is found that the cooperative execution of SA and distribution alignment has obvious advantages, which can effectively reduce spectral shift and better complete the cross-scene HSI classification task.

In addition, computational complexity of each algorithm is evaluated. All the experiments are carried out using MATLAB on an Intel(R) Core(TM) i7-9750H Central processing unit with 16GB of RAM. The total time of training and testing of each method on the Houston data set is listed in Table VII, and the number of training samples in the SD are the same as in the previous experiments. In Table VII, MEDA takes the longest time, which is based on the kernel strategy. It has a high computational cost, and may not be suitable

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TABLE VII
EXECUTION TIME (IN SECONDS) OF DIFFERENT METHODS IN THE HOUSTON DATA SET

Method	*Times(s)*
TCA	7.5
CORAL	1.4
STL	10.2
SCA	53.5
JDA	20.2
JGSA	94.5
MEDA	524.1
DSA	97.7
DCA	115.3

for handling large-scale data. The proposed DCA is iterative, so the calculation time is longer than state-of-the-art TCA, CORAL, and STL without iteration.

V. CONCLUSION

In this article, an efficient DCA has been proposed for improving the traditional domain adaptive methods and reducing the spectral shift in cross-scene HSI classification. SA and distribution alignment work cooperatively, in which the subspaces of SD and TD are learned under the subspace correlation constraint, and then the marginal distribution and conditional distribution are considered for alignment using the MMD is used for distribution alignment. In addition, a data reconstruction constraint is applied to SD and TD to maintain the discrimination information in the original samples. Comprehensive experiments on one synthetic data and three real hyperspectral data have verified the effectiveness of the proposed DCA in reducing domain shift and significantly improving cross-scene HSI classification performance.

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